

Design of Optical Systems for Hybrid Photoluminescence Imaging, Spectrometer and Dynamic Moiré Displacement Sensor

Submitted by
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PALAKKAD

May 10, 2026

Works Done

BTP

BTP

- Hybrid Photoluminescence Imaging System

BTP

- Hybrid Photoluminescence Imaging System
- Design of Spectrometer

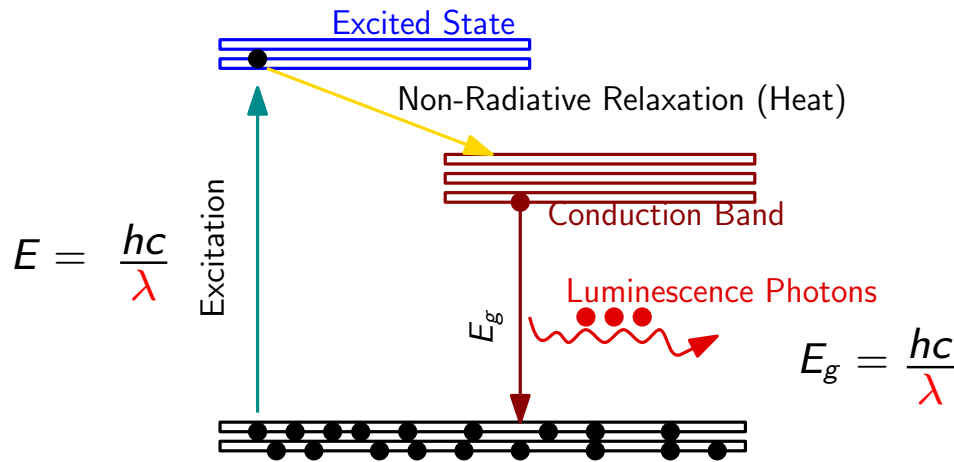
BTP

- Hybrid Photoluminescence Imaging System
- Design of Spectrometer
- Dynamic Moiré Displacement Sensor

Photoluminescence of Solar Cells - Overview

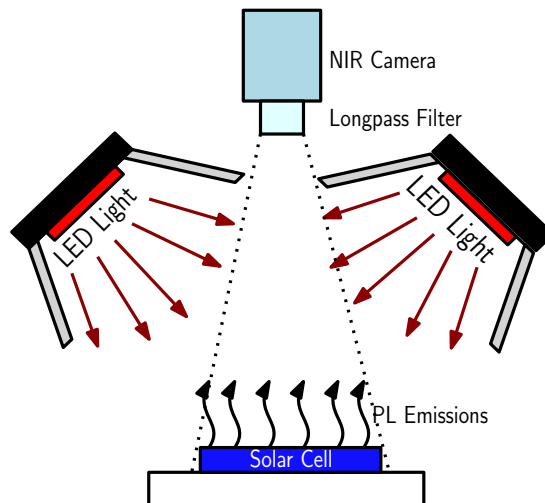
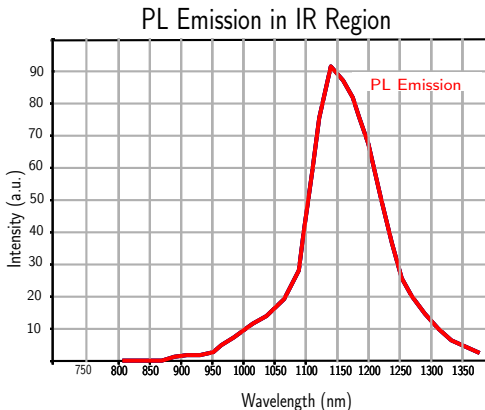
Luminescence

Luminescence is a phenomenon in which, excited electrons in solar cells release their energy as **photons** through **radiative recombination**

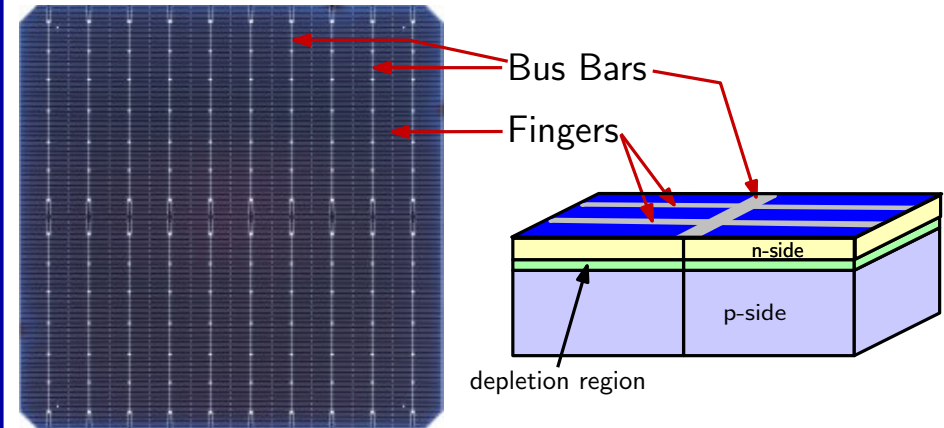


Photoluminescence Imaging for Solar Cells

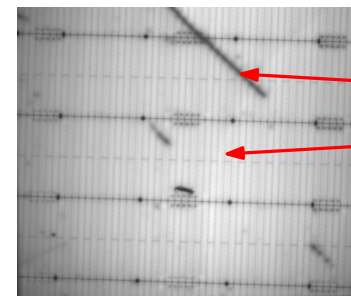
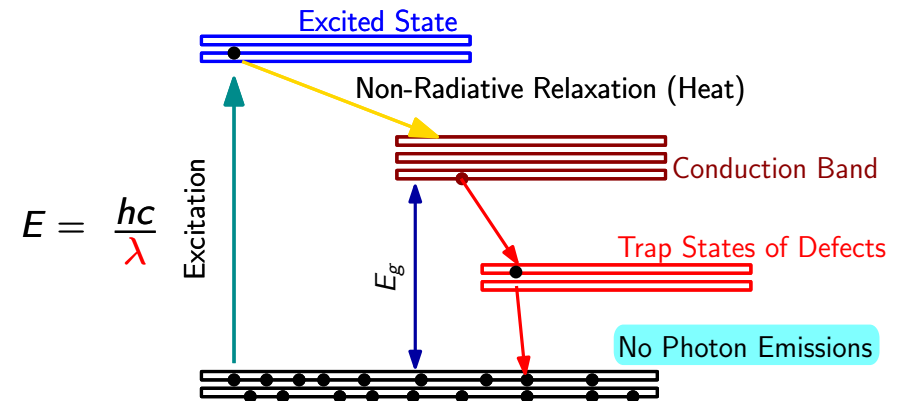
$$I_{PL} = A G \tau$$



Solar Cells



Shockley-Read-Hall Recombination due to Defects



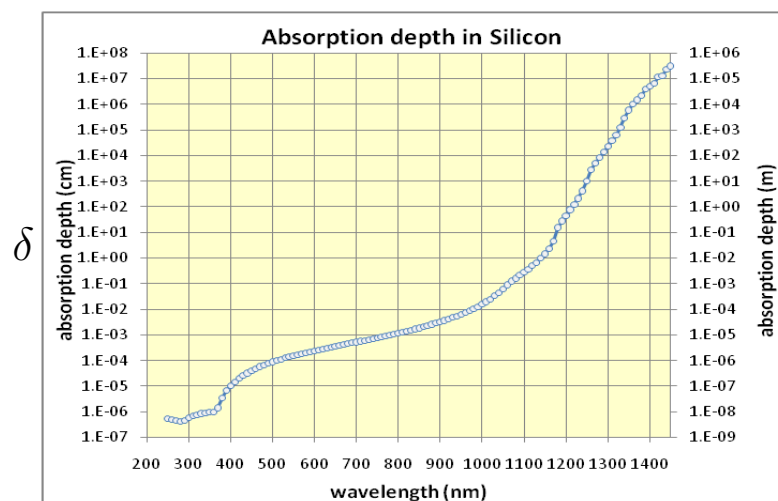
PL Image

Challenges in Photoluminescence Measurements

Depth of Penetration

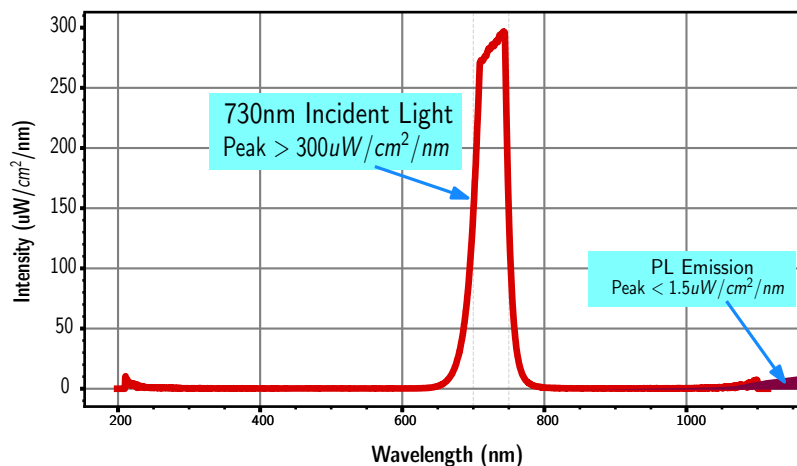
Beer-Lambert Law

$$I(z) \approx I_0 e^{-\alpha z} \quad \text{Absorption Depth, } \delta = \frac{1}{\alpha}$$



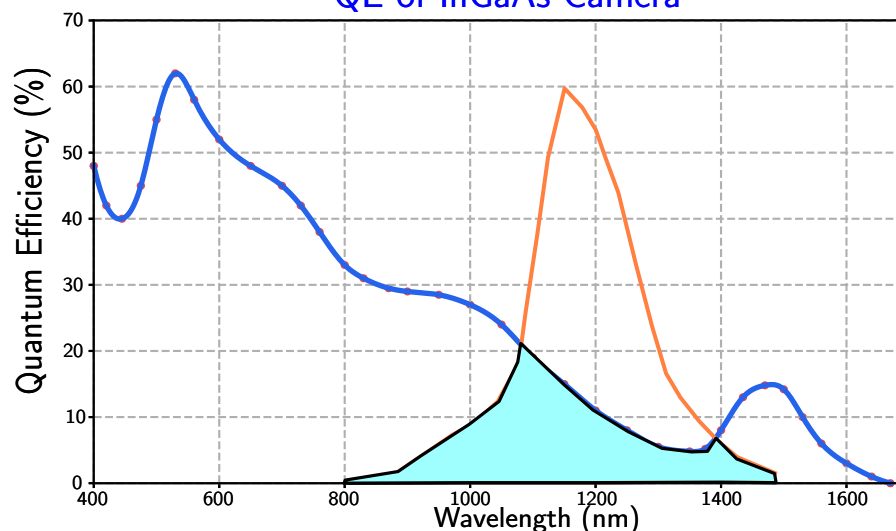
Incident 730nm LED

Incident Light and PL Emission

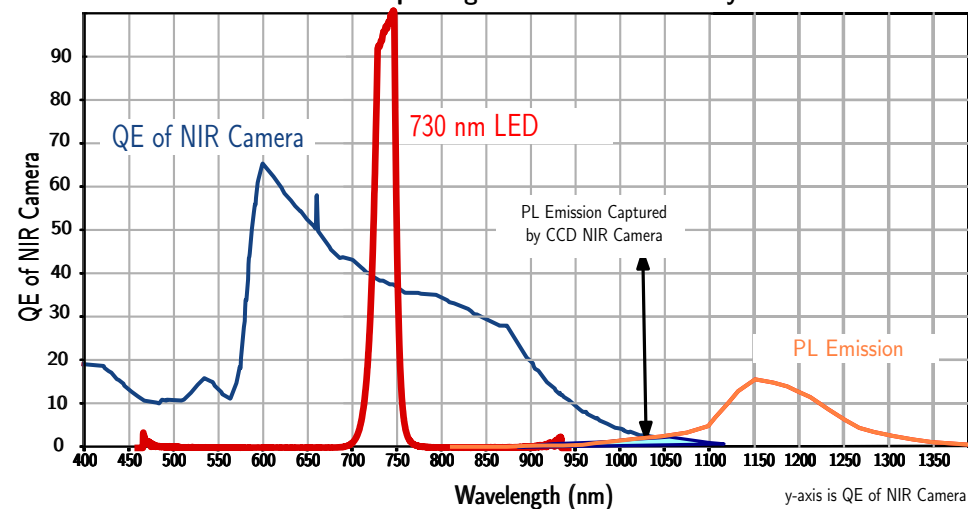


Camera for Luminescence Capture

QE of InGaAs Camera



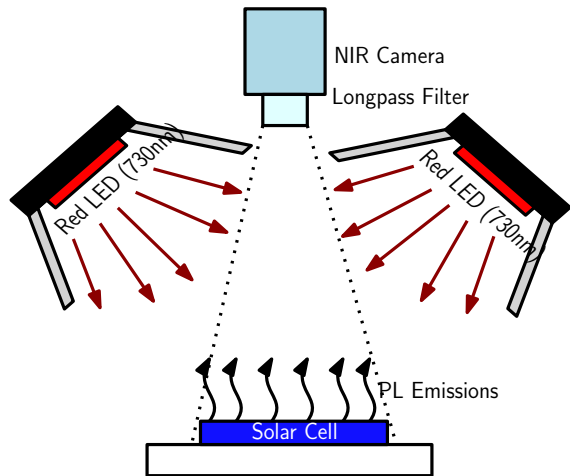
Limitation of Capturing Photoluminescence by CCD NIR Camera



We Should Filter Incident Light Completely to Capture the weak PL Emissions

Optical Filtering of Incident Light

Photoluminescence Imaging System



Parameters of Optical Filter

Optical Density (OD)

How much of the incident light will be blocked by the filter in each wavelength

$$OD = -\log_{10}(\text{Transmittance})$$

$$\text{Transmittance} = 10^{-OD} * 100\%$$

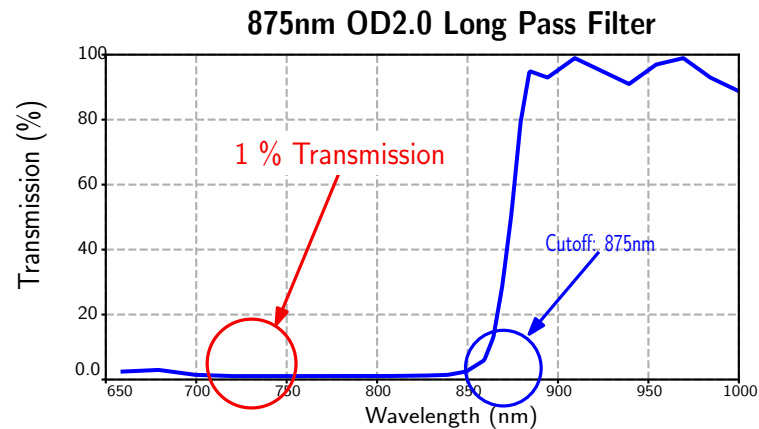
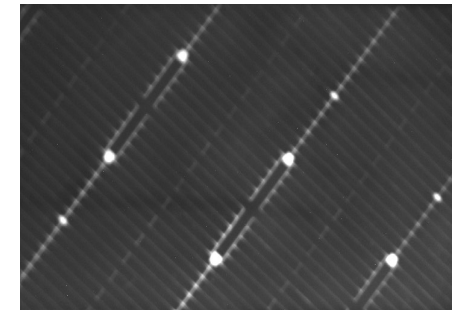
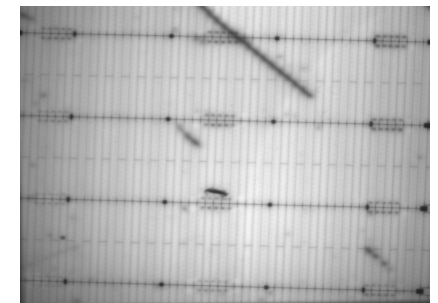
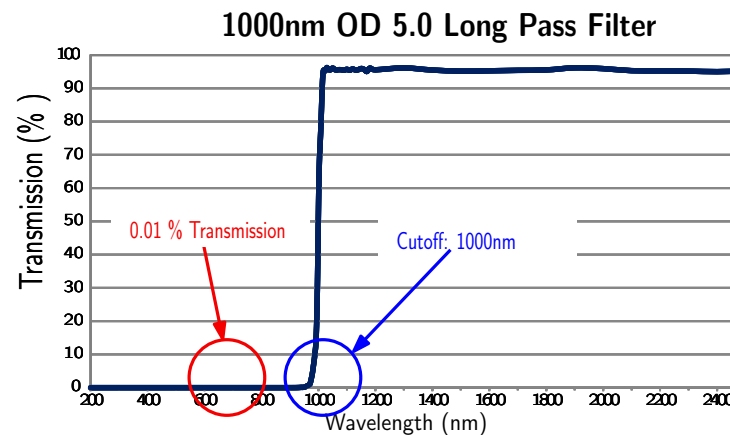


Image from NIR Camera

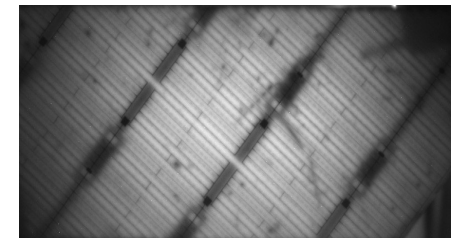
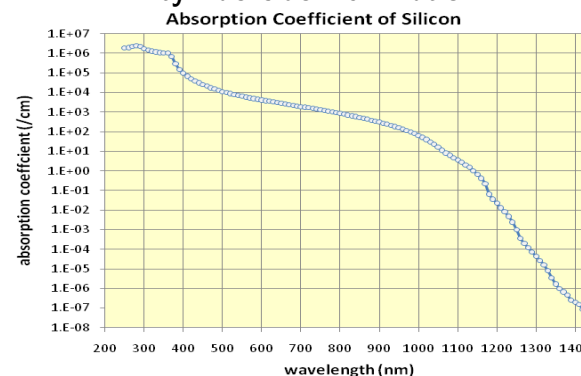


With OD 2.0 Filter



With OD 5.0 Filter

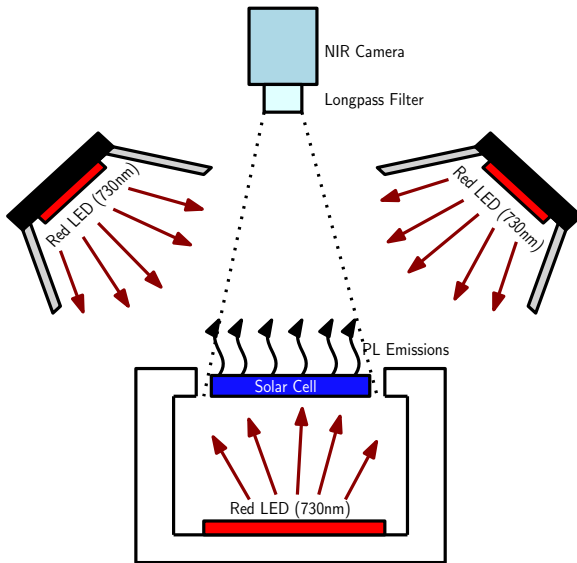
Inherent High Pass Filtering by Backside Illumination



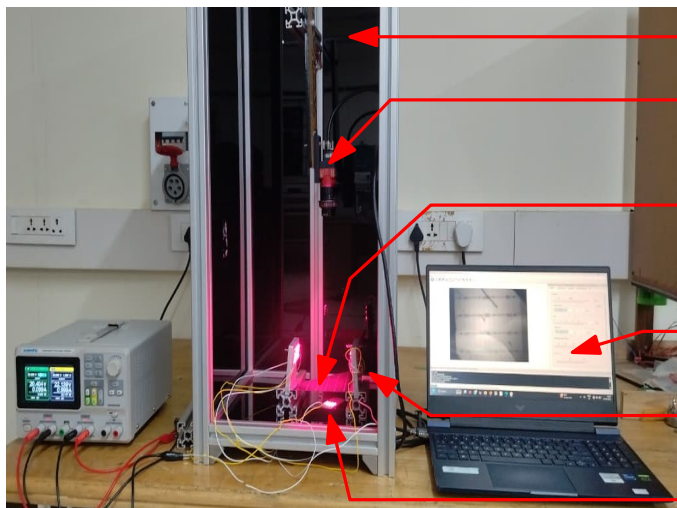
With Backside Illumination
Inherent Filtering

Hybrid Photoluminescence - Top and Bottom Illumination

Hybrid Photoluminescence Imaging System



Hybrid PL Imaging - First Prototype



Dark Box

Allied Vision
NIR Camera

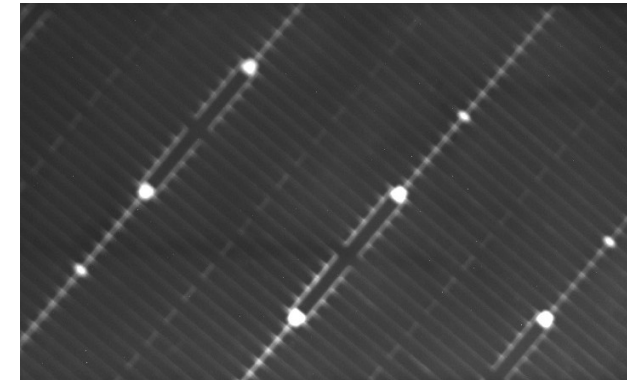
Solar Cell

Camera Interface

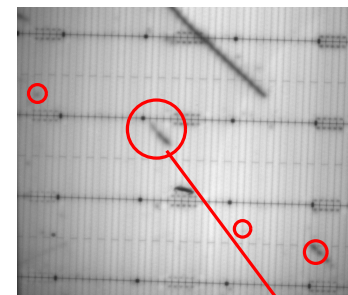
Top Illumination

Back Illumination

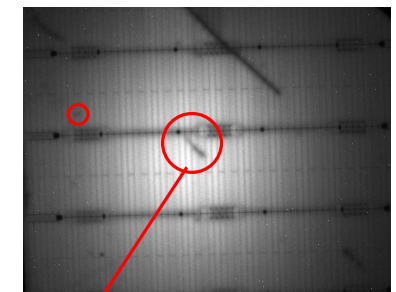
Results



PL Emissions not captured as
Reflection of incident light dominates
without Filter



Top Illumination
OD5.0 Filter



Bottom Illumination
With OD5.0 Filter

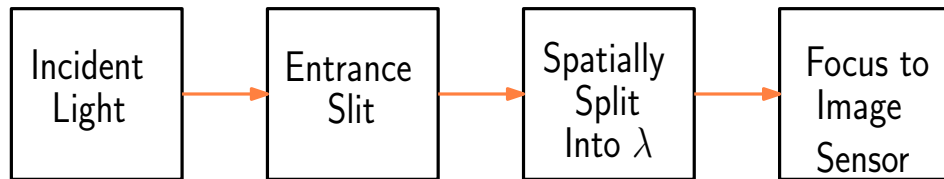


Defective Parts where PL Emissions are not happening

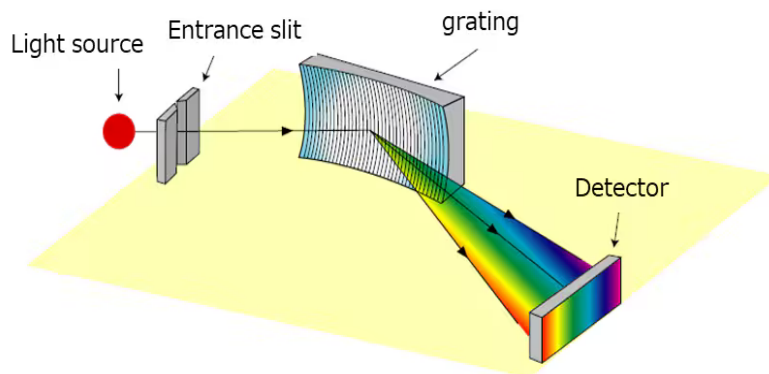
Spectrometer - Overview

What is Spectrometer

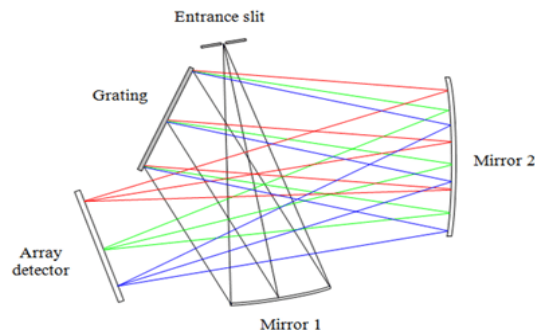
Device used to characterize properties of light with respect to it's wavelength. Eg; Reflectance, Absorbance, Transmittance etc...



Concave Grating



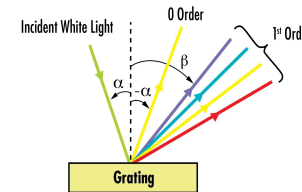
Crossed Czerny-Turner



Components

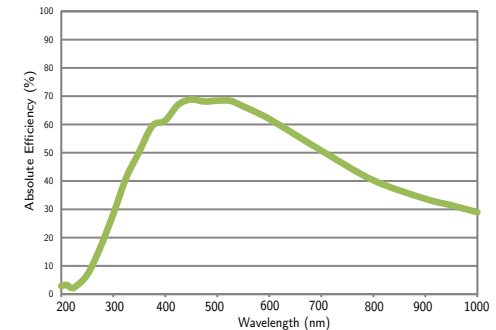
Entrance Slit: regulates the amount of light entering system and controlling the spectral resolution

Diffraction Grating: Spatially splits incident light into different wavelength components



$$d(\sin \alpha + \sin \beta) = m\lambda$$

$$d = \frac{1}{\text{Groove Density}}$$



Focusing Optics: Focuses each wavelengths onto image sensor pixels
Should consider the efficiency (λ)

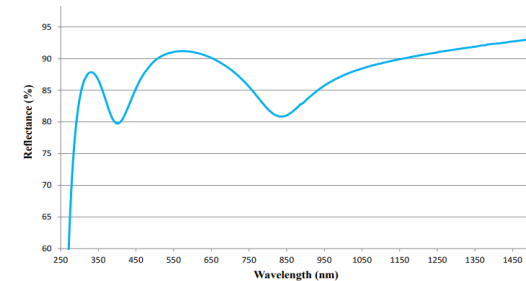
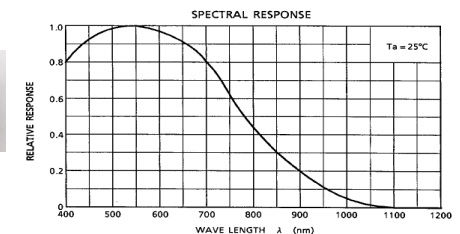
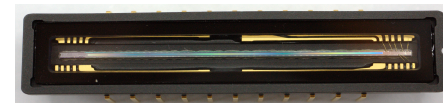
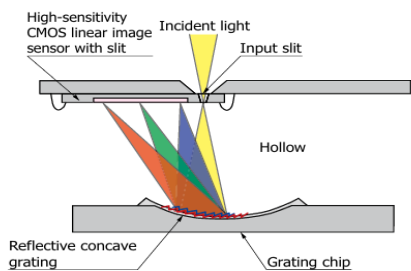


Image Sensor: Captures intensity of light for different wavelengths
Should consider the QE and pixels to determine resolution of system



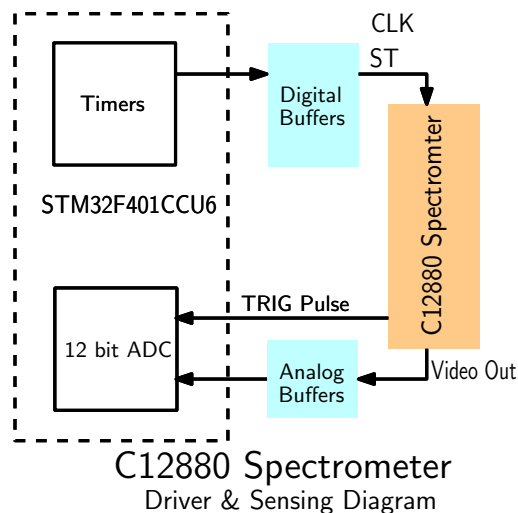
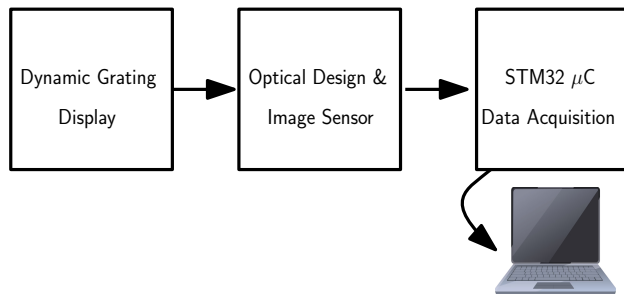
C12880 Spectrometer - Hamamatsu

Internal Structure of C12880



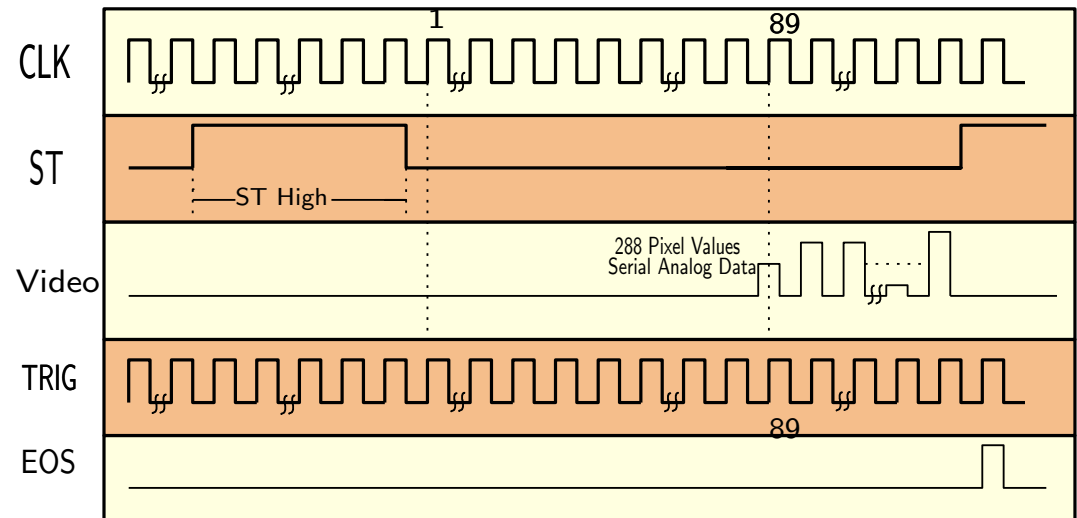
288 Pixels

- Spectral Range: 340 nm- 850nm
- Spectral Resolution: 15nm (datasheet)

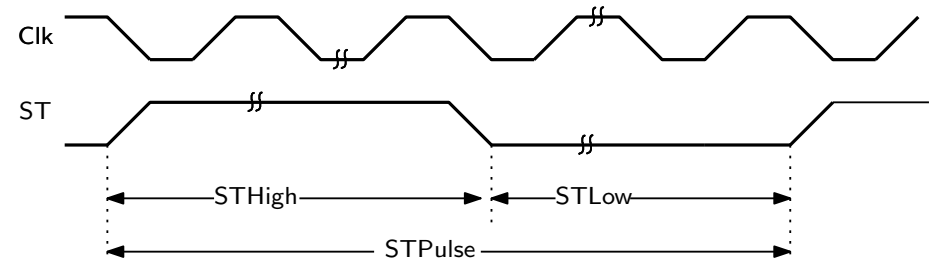


Timing Diagram

$$T_{int} = ST_{High} + 48 T_{Clk}$$



Timing Constraints



Parameter	Min (s)
STPulse	$\frac{381}{f(CLK)}$
STHigh	$\frac{6}{f(CLK)}$
STLow	$\frac{375}{f(CLK)}$

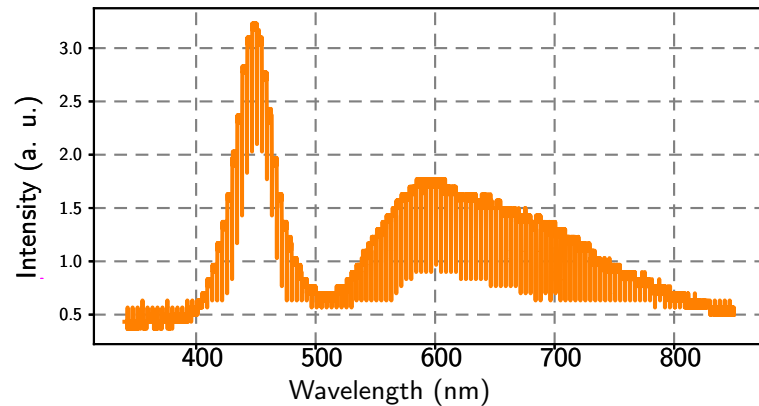
$$Integration\ Time = ST_{High} + 48 T_{Clk}$$

C12880 Spectrometer - Results & Comparison

C12880 Mini-Spectrometer



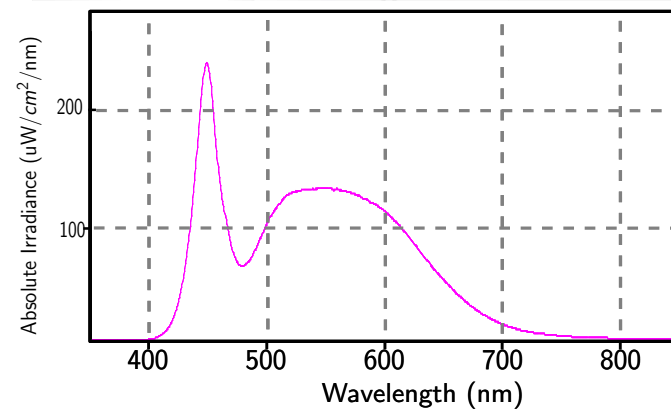
Spectrum from C12880 Spectrometer



Ocean Optics Spectrometer

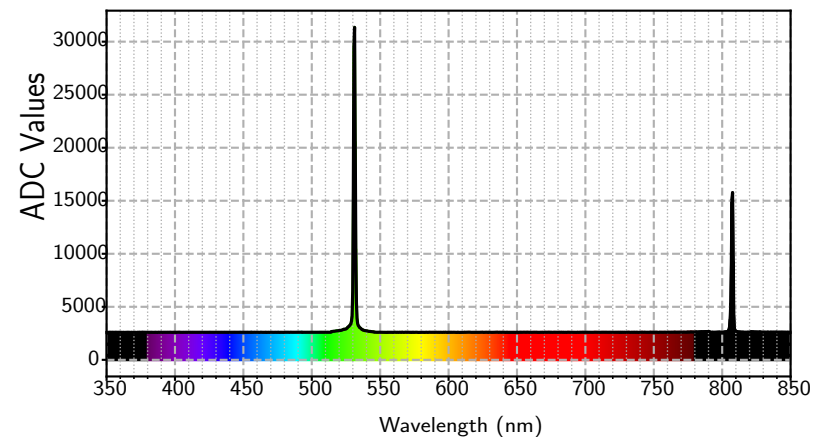
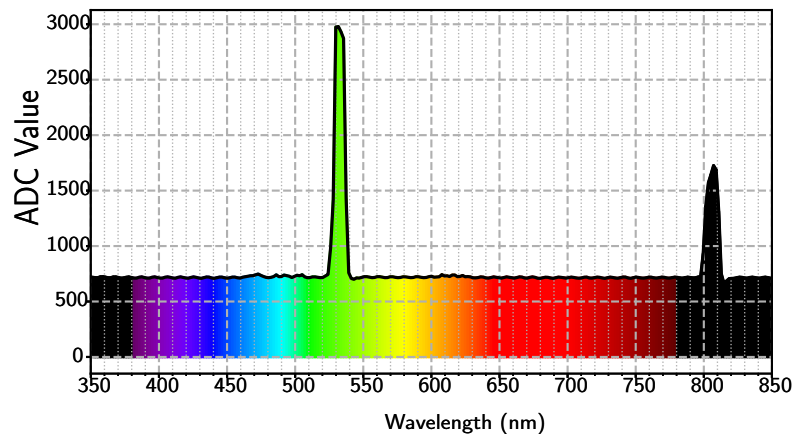


Spectrum from Ocean Optics Spectrometer



White Light Source

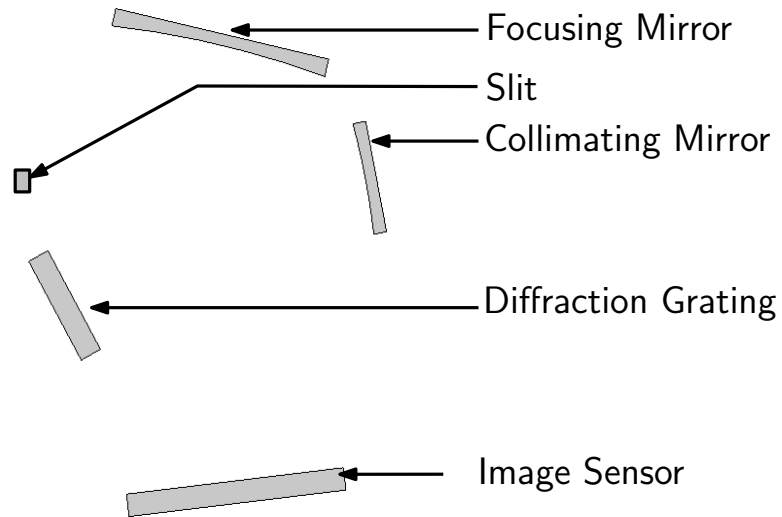
Light Source with
peaks at 532nm & 807nm



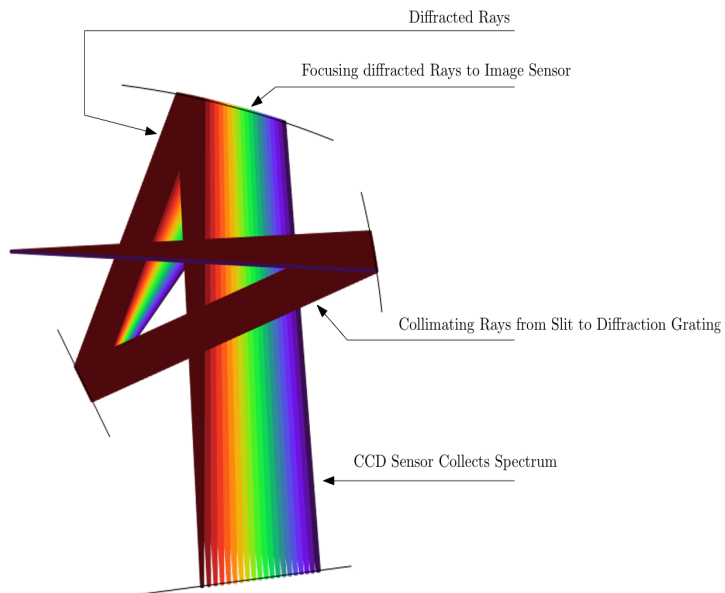
- Spectral Range: 340 nm- 850nm
- Spectral Resolution Achieved (FWHM) : 20 nm

Czerny-Turner Spectrometer: Optical Design in COMSOL

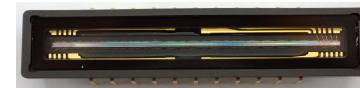
Czerny-Turner Configuration



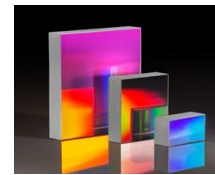
Ray Tracing in COMSOL Multiphysics



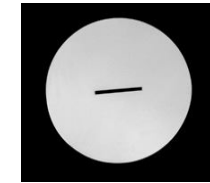
Components and Design



TCD1304 Image Sensor
3648 pixels



600 Grooves/mm
Diffraction Grating

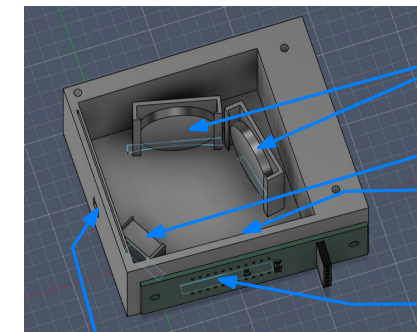


Entrance Slit (5-10 μ m)



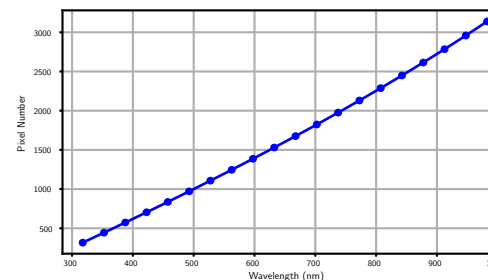
50mm FL/50mm dia
Concave Mirror

3D Design of Spectrometer

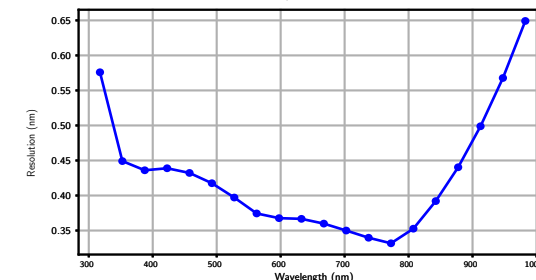


Pixel Map and Resolution wrt λ

Pixels to Wavelength Mapping



Resolution v/s Wavelength Mapping

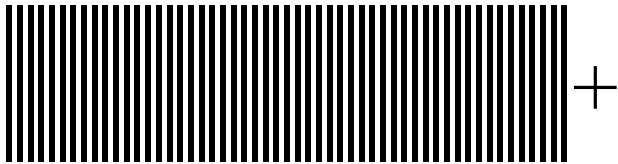


Moire' Effect - Overview

- Two or more periodic patterns with similar but slightly different frequencies or phases overlap to form Moire' Pattern
- Resultant Pattern is formed a result of the constructive and destructive interference of two base patterns
- Moire' pattern is directly related to the type of mismatch between the patterns (E.g. frequency, angle) and it can be back calculated

Moire' Effect - Overview

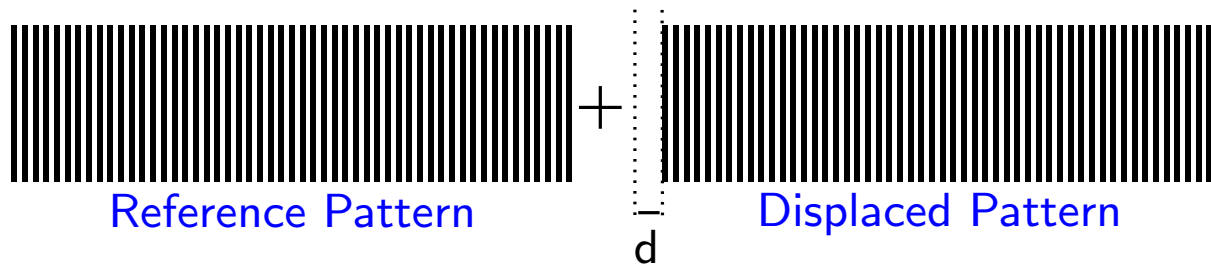
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Reference Pattern

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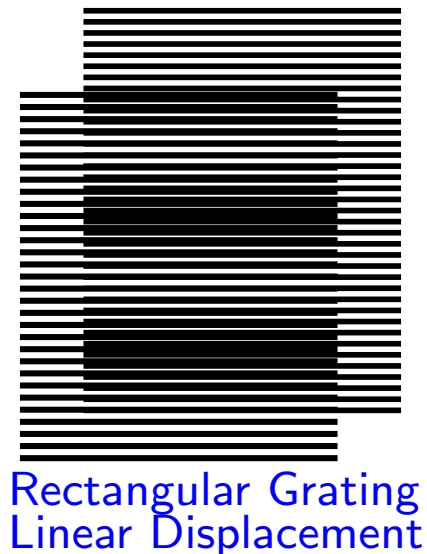
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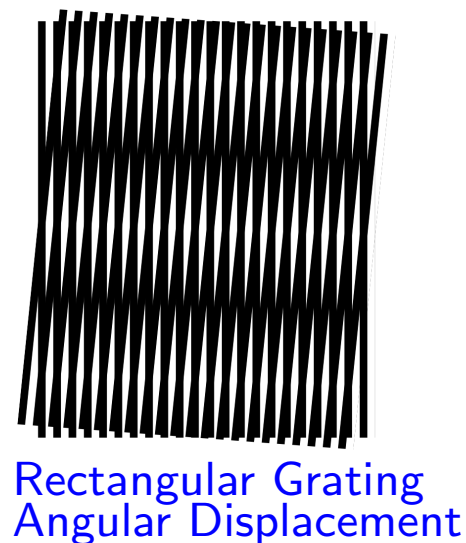
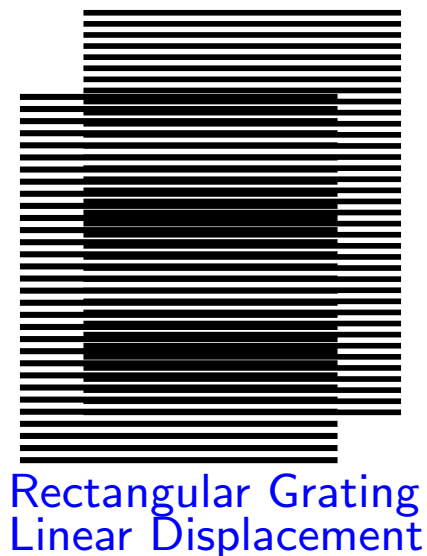
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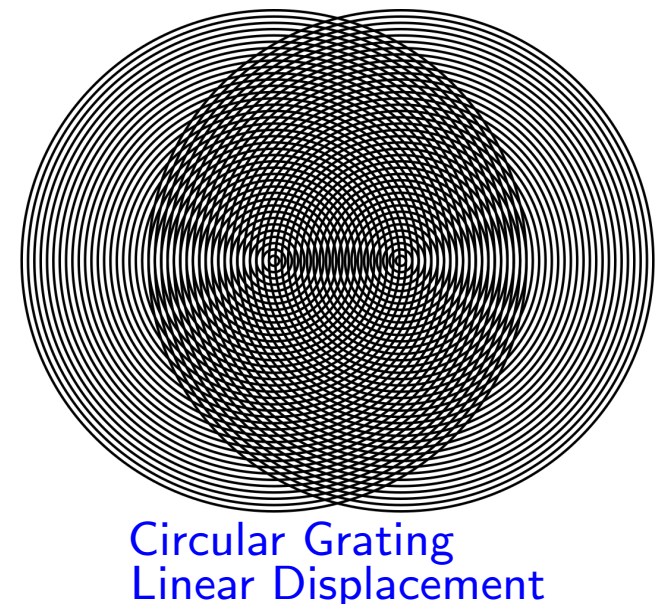
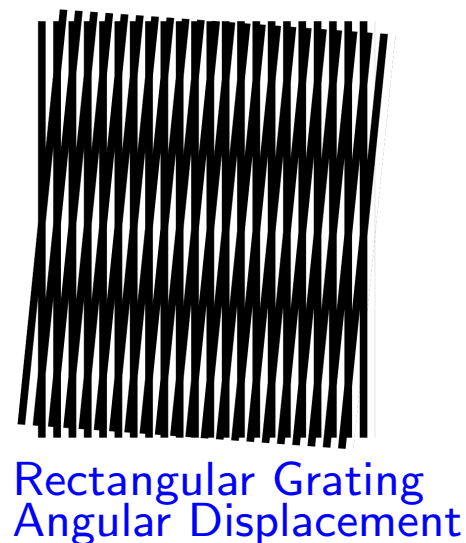
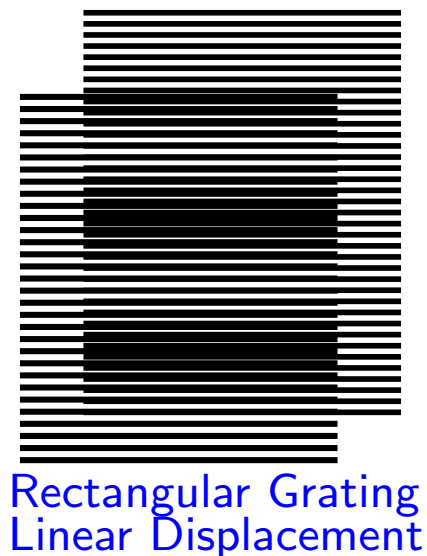
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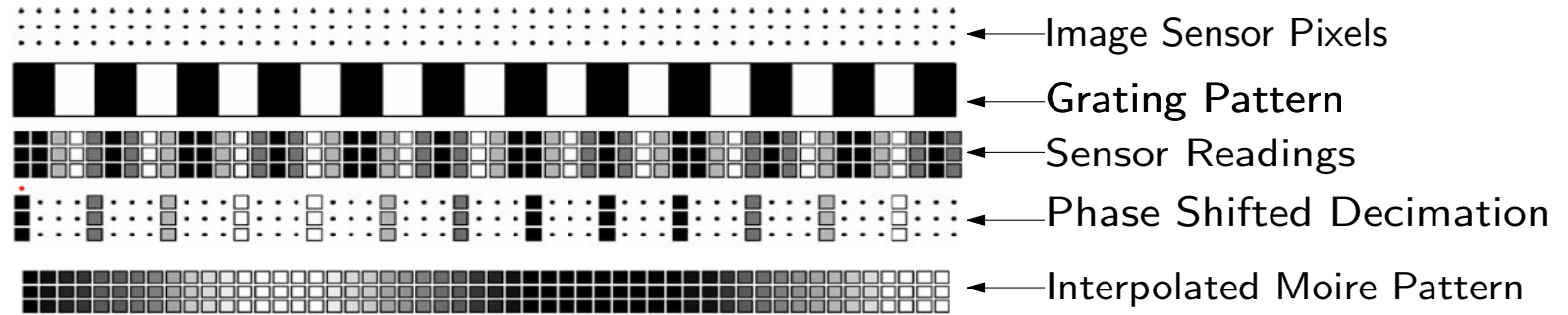


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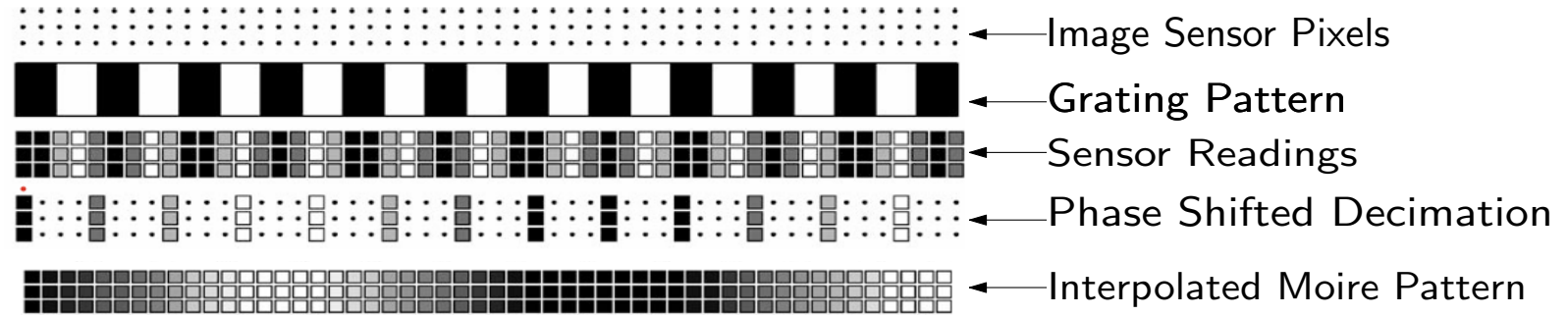


Sampling Moire Method [2] - Overview



Sampling Moire Method [2]

Sampling Moire Method [2] - Overview



Sampling Moire Method [2]

Mathematical Modelling [2]

$$I_k(i) = I_a(i) \cos \left(\phi_m(i) + \frac{2\pi k}{N} \right) + I_b(i)$$

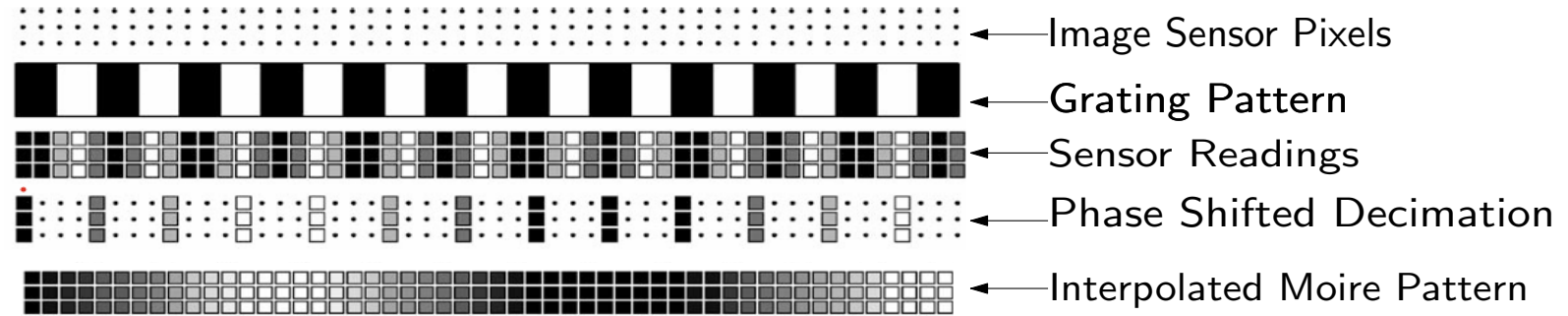
$$\phi_m(i) = -\tan^{-1} \left(\frac{\sum_{k=0}^{N-1} I_k(i) \sin \left(\frac{k2\pi}{N} \right)}{\sum_{k=0}^{N-1} I_k(i) \cos \left(\frac{k2\pi}{N} \right)} \right)$$

$$\Delta\phi_M = \phi_{def} - \phi_{ref}$$

$$\text{Displacement } u = p \cdot \frac{\Delta\phi_M}{2\pi}$$

p = pitch of grating

Sampling Moire Method [2] - Overview



Sampling Moire Method [2]

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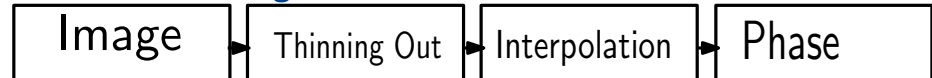
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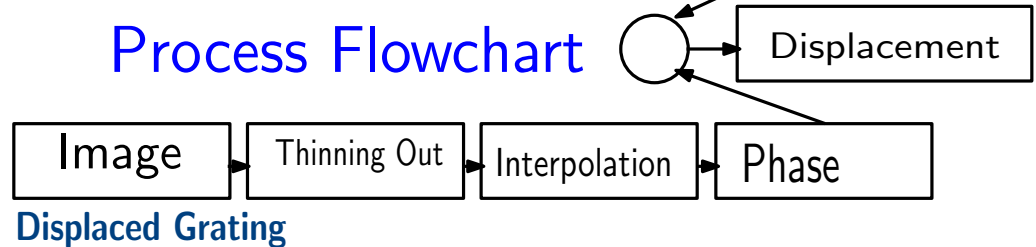
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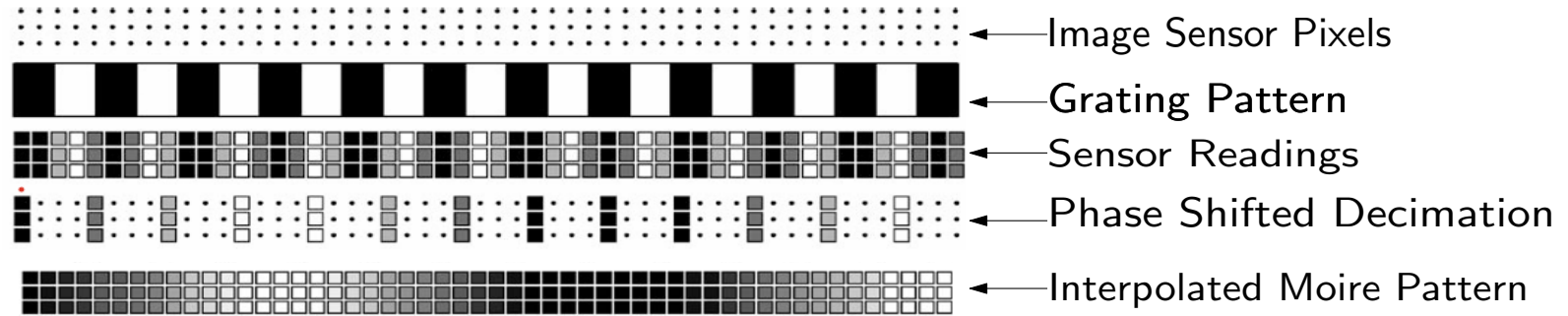
Reference Grating



Process Flowchart



Sampling Moire Method [2] - Overview



Sampling Moire Method [2]

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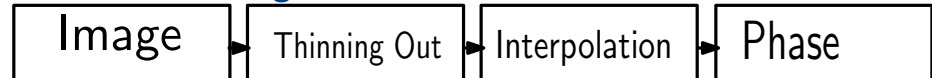
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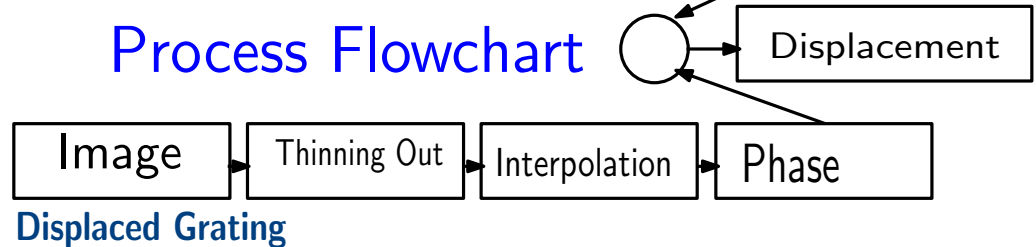
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Reference Grating



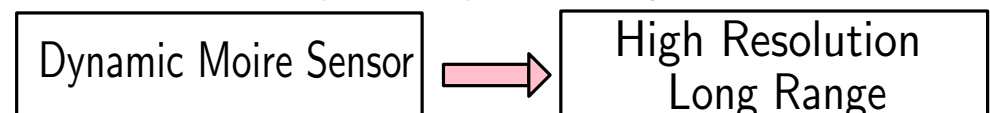
Process Flowchart



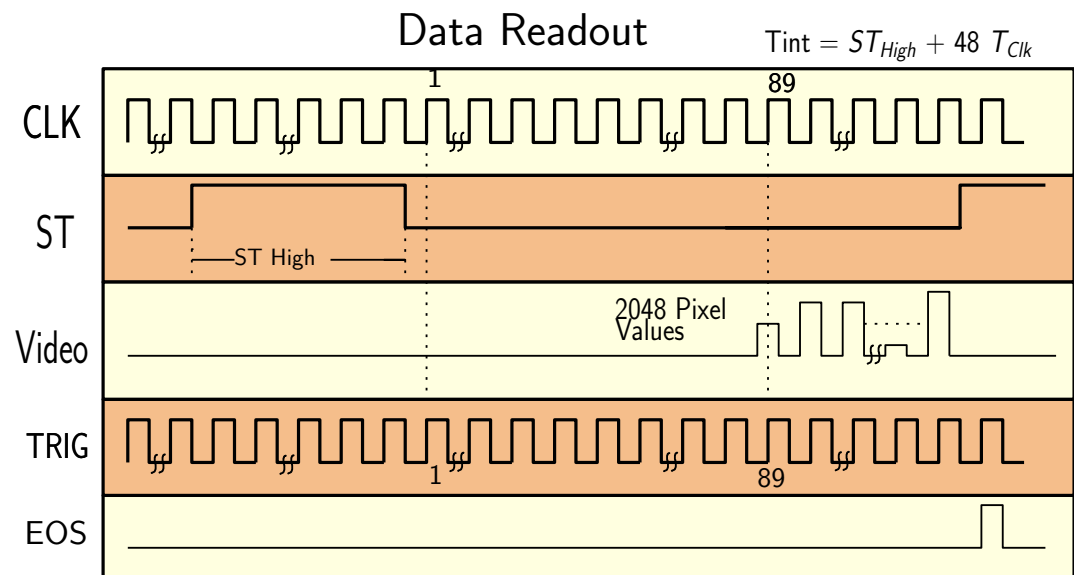
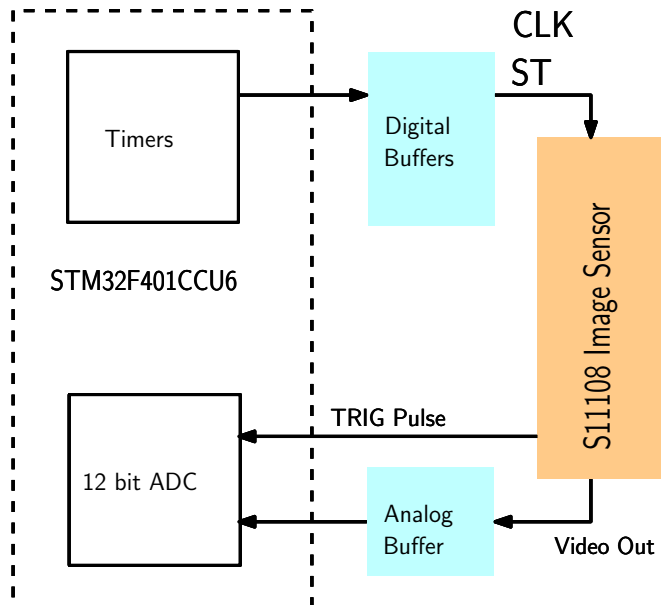
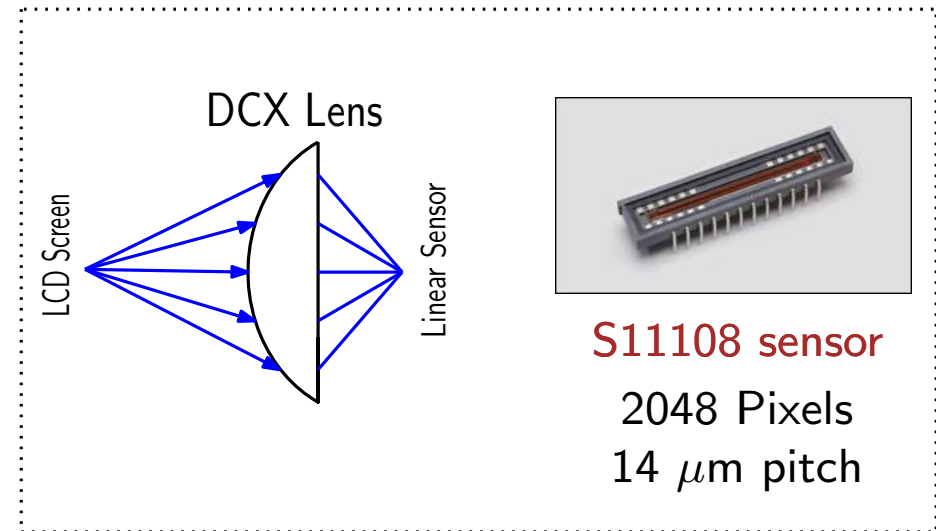
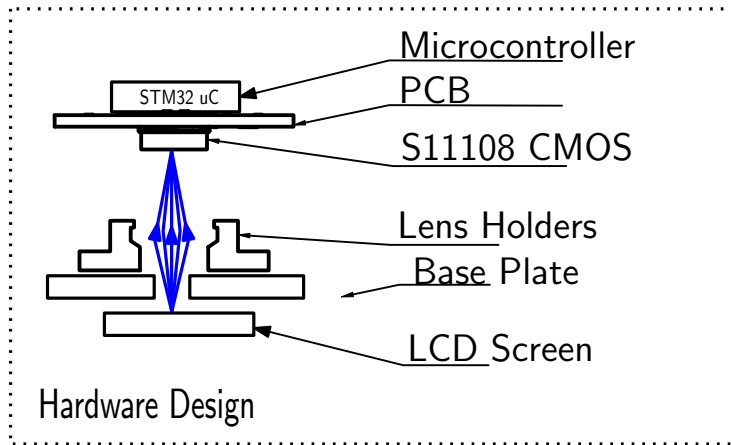
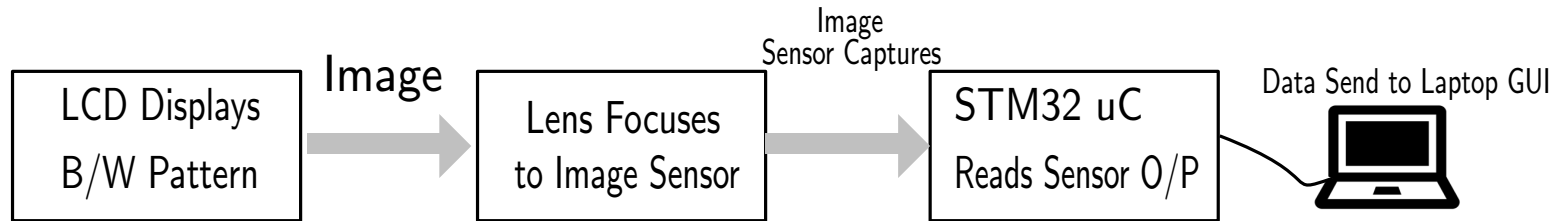
$$u = p \cdot \frac{\Delta\phi_M}{2\pi}$$

Range = 1 pitch Resolution $\approx \frac{1}{1000}$ pitch

Dynamically Controlling Pitch

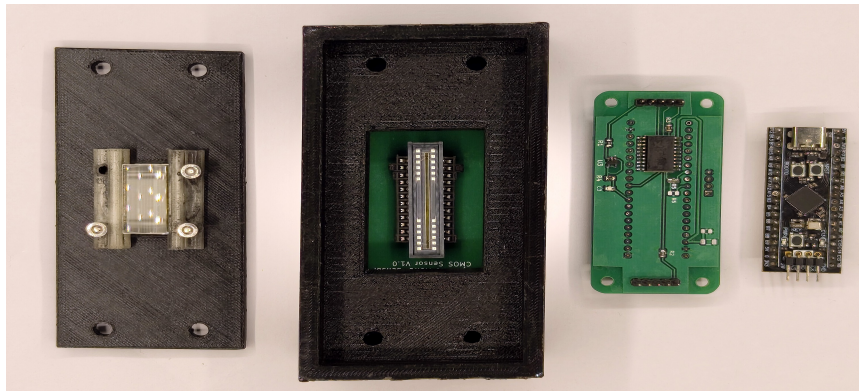


Dynamic Moire' Sensor : Hardware Design - I



Dynamic Moire' Sensor : Hardware Design - II

Components

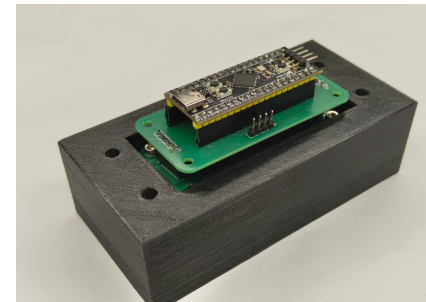


Lens Holder

Case

PCB & STM32 uC

Test - 1



Moire' Sensor V1.0

+



2mm Transparent Grating

Assembled Dynamic Moire Hardware

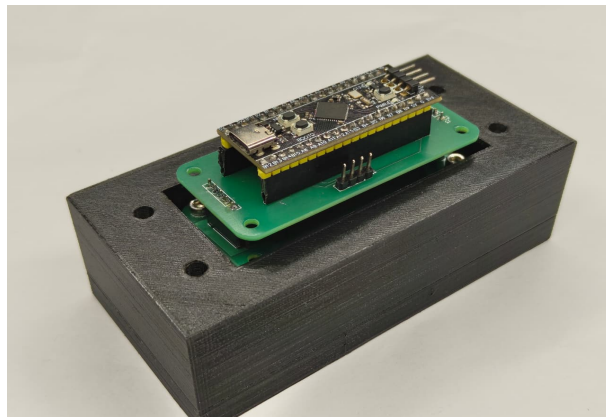
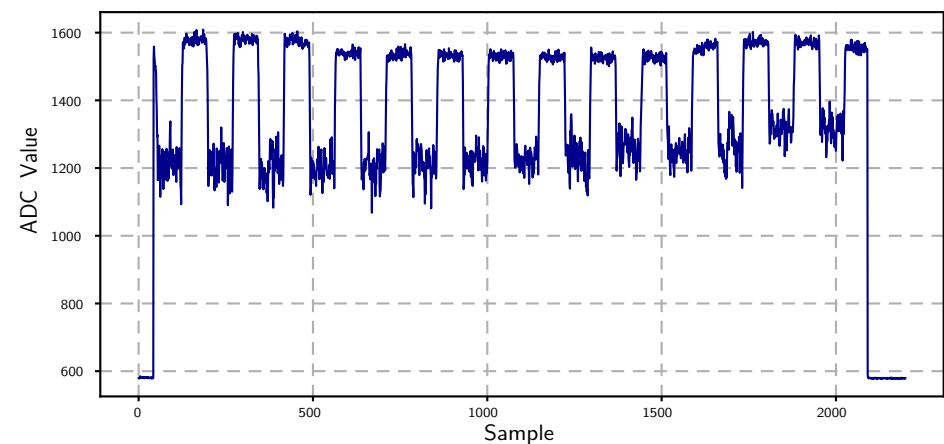
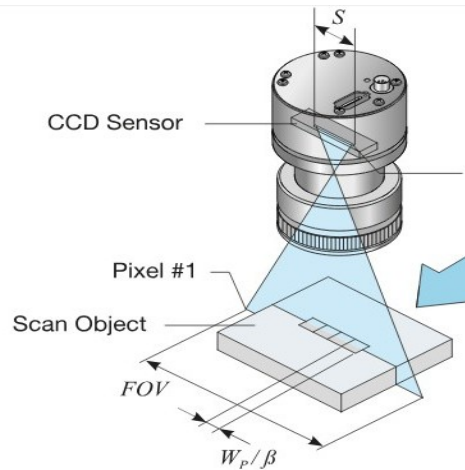


Image of 2mm Pattern

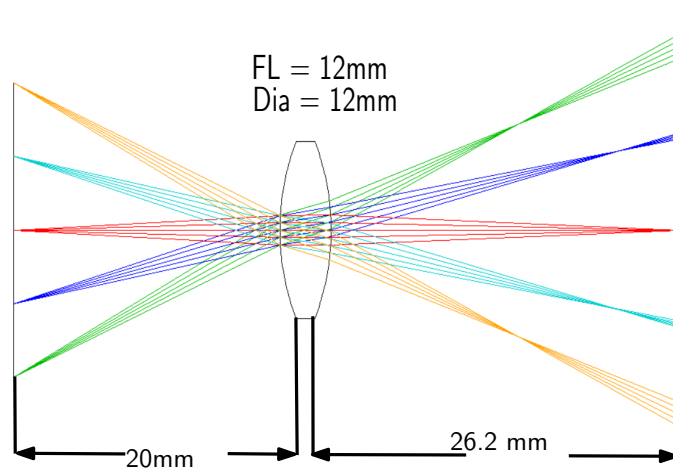


Optical Design and in OSLO

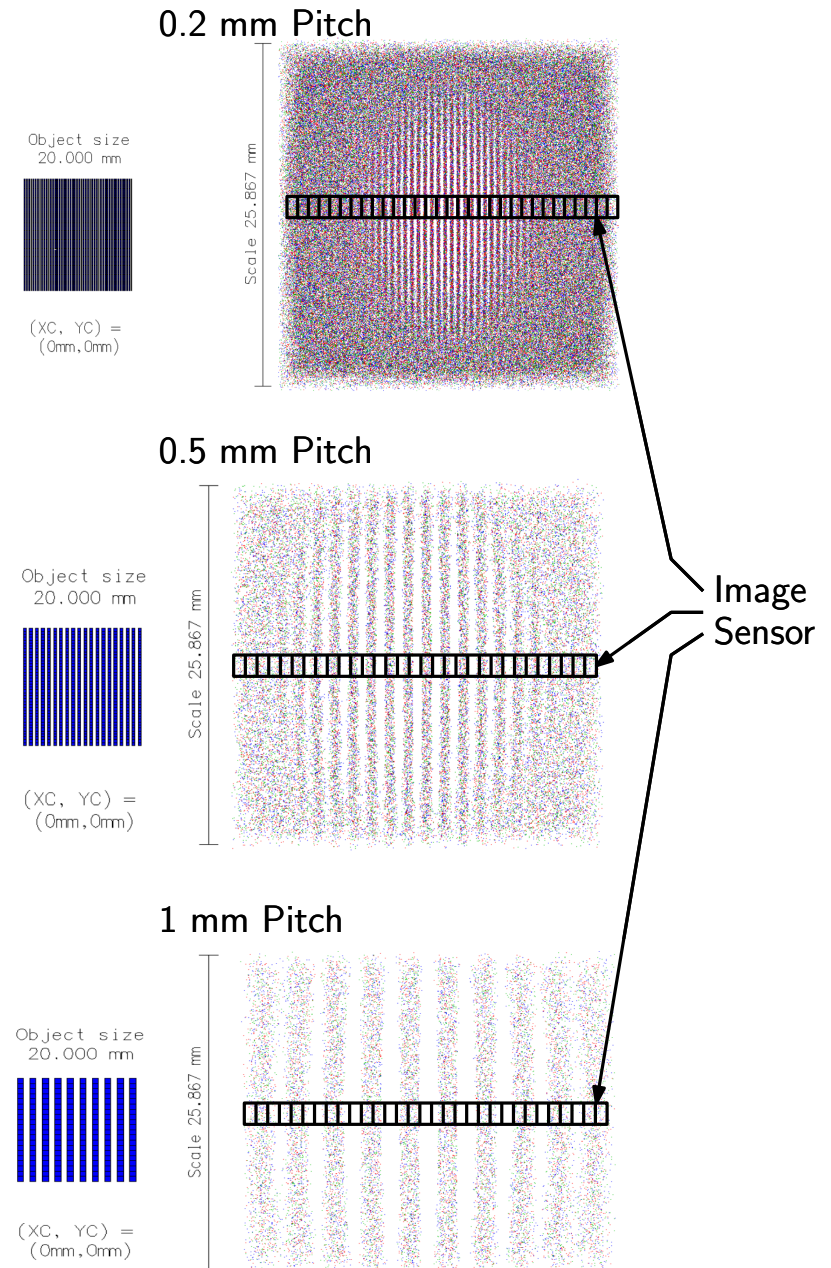
Line Scan Cameras



Optics Simulation in OSLO

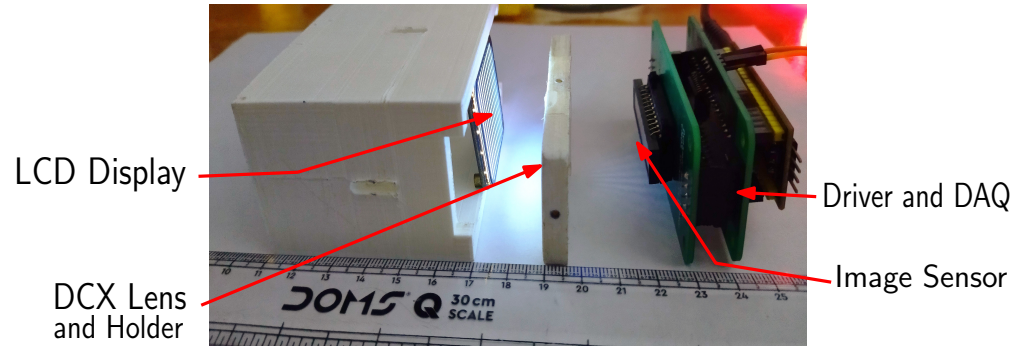


OSLO Simulation with Different Pitch

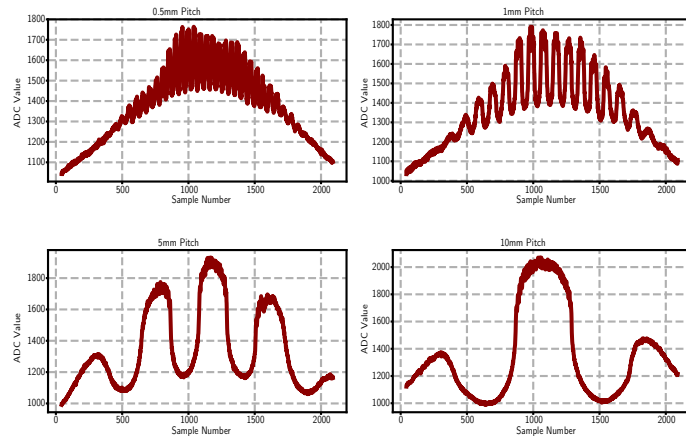


Dynamic Moire' Sensor - Test Setup & Results

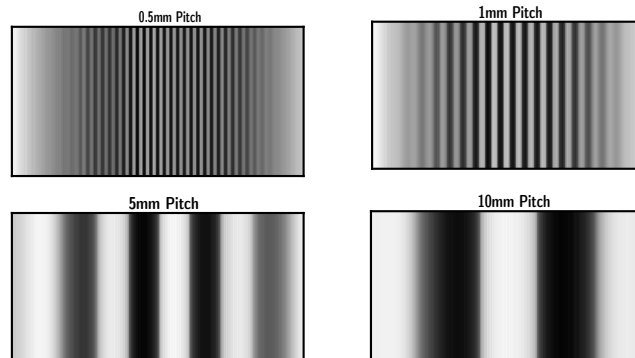
Optical Design Verification



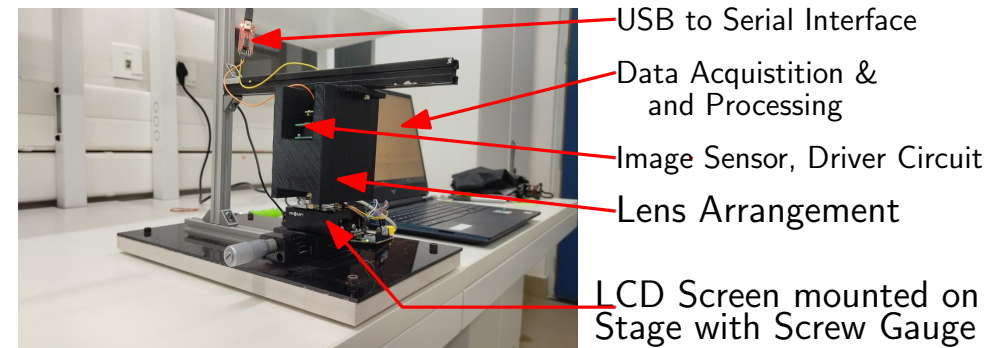
ADC Reading



Images with Different Pitch

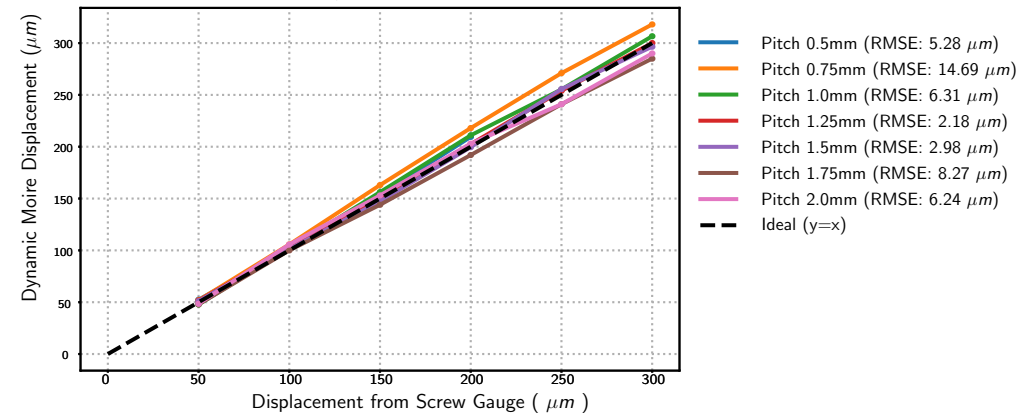


Displacement Measurement



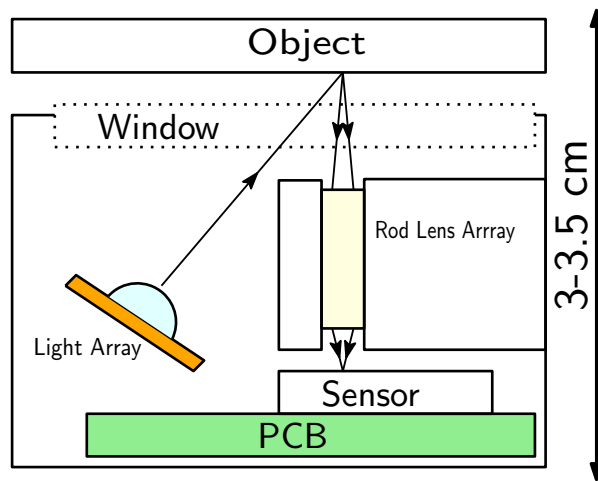
Results

Moire Displacement v/s Screw Gauge Displacement



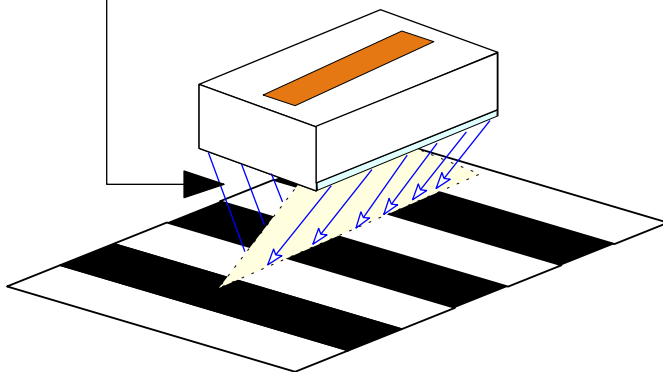
Dynamic Moire' Sensor - Future Works

Rod Lens Array for Imaging



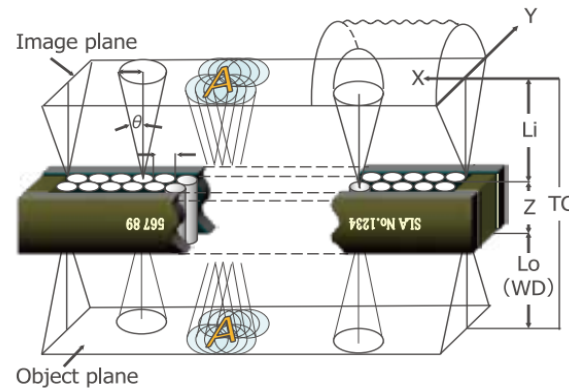
CIS Module

- Contact Image Sensor
- LED Array Design

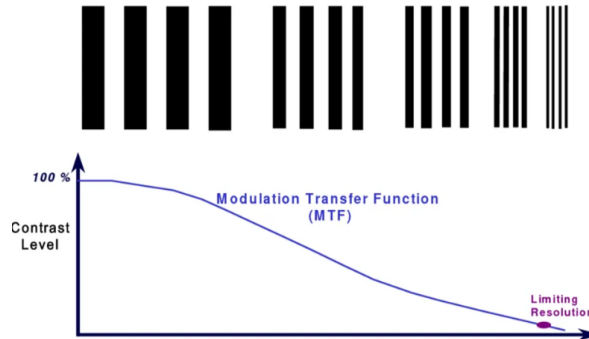


E-Grating Display

Rod Lens Array in CIS Module



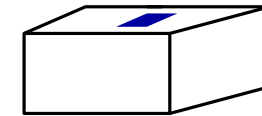
1:1 Imaging with higher MTF
80% at 300 dpi



$$\% \text{ Contrast} = \left(\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \right) \times 100\%$$

Dynamic Moire' X-Y Stage

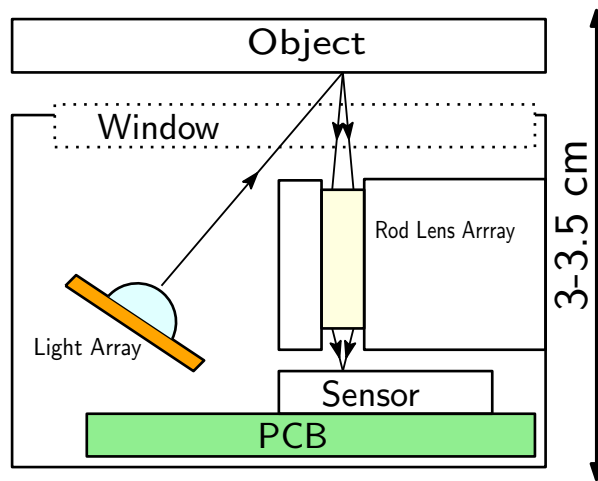
Image Sensor



E-Grating Display
Dynamically Changing pitch
and X-Y Direction

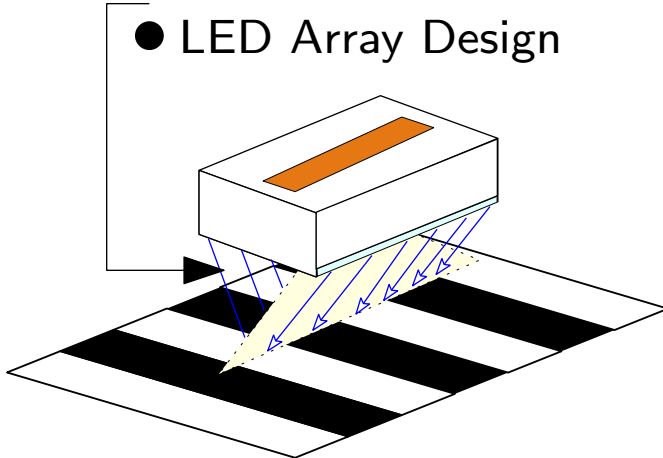
Dynamic Moire' Sensor - Future Works

Rod Lens Array for Imaging



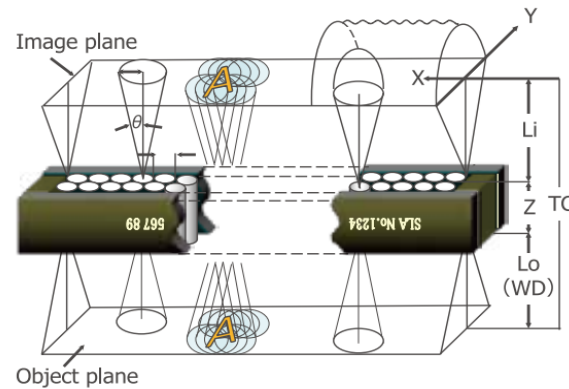
CIS Module

- Contact Image Sensor
- LED Array Design

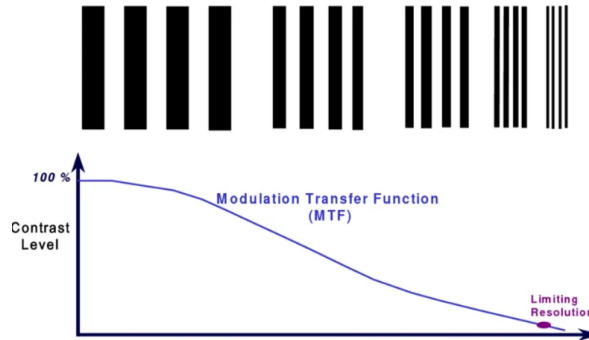


E-Grating Display

Rod Lens Array in CIS Module



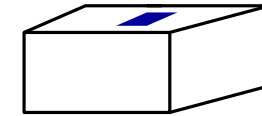
1:1 Imaging with higher MTF
80% at 300 dpi



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Dynamic Moire' X-Y Stage

Image Sensor



E-Grating Display
Dynamically Changing pitch
and X-Y Direction

Outcomes of Projects

- **Hybrid Photoluminescence System Design**
 - R&D Completed, awaiting commercial production
- **Design of Spectrometer**
 - C12880 Spectrometer: system design completed and is used for other research projects
 - Czerny Turner Spectrometer: Optical design finalized, prototyping in progress
- **Dynamic Moiré Sensor**
 - Patenting application in preparation

Acknowledgement

Dr Arvind Ajoy - BTP Guide

- Ananthu Sasikumar - Mentorship in Dynamic Moire Project
- Abhinav Rajeev - Mentor for Spectrometer and Hybrid Photoluminescence Imaging
- CSquare Lab and C-06 : Laboratory facilities

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Thank You :)